

Simplified Research Abstracts with Contribution Tagging

TRA301: Final Project

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1. Introduction

Non-expert readers, such as high-school students or multidisciplinary collaborators, and the general public, may struggle to identify actual advances in scientific abstracts due to their rich mix of technical jargon and innovative results. Meanwhile, most automatic summarization systems regard the abstract as a single block of text, therefore masking the “contribution” assertions that represent true scientific advancement. In this paper, we handle these issues by creating a pipeline that first labels each sentence in an abstract as either <BACKGROUND> or <CONTRIBUTION> and then produces a clear high-school level summary that focuses solely on the tagged contribution statements. To enable these processes, we created a corpus of 7,000 arXiv abstracts associated with their most significant references. From there, we manually annotate a representative fraction for training, and adjust our simplification module to achieve Flesch Reading Ease scores of 50-70. By separating “why it matters” from “what was done,” our approach not only sharpens focus on the paper’s innovations, but also provides a fine-grained supervision for both tagging and summarization. Ultimately, this makes cutting-edge research more transparent, pedagogically useful and widely accessible.

2. Background and Related Work

Significant contributions are frequently hidden in scientific publications buried with technical jargon. This renders them unreadable by the general public and there have been prior efforts to overcome this obstacle in a number of ways. For example, Liu et al. present “ContributionSum” which aimed to be a large-scale dataset and generation framework aimed at producing disentangled contribution summaries for computer science research papers [9]. Unlike general summarization systems that focus on abstracts or full-text compression, ContributionSum specifically targets author-listed

contributions and classifies them into structured categories like “Approach,” “Analysis,” “Result,” and others. When evaluating on ROUGE and BERTScore metrics, their summarization efforts perform better than mainstream summary baselines.

Separately, Jurgens et al. developed an NLP contribution graph that created organized citation graphs of novel ideas by mining and connecting explicit scientific contributions from computer science journals [6]. This approach provides an idea-centric perspective of scientific advancement and allows for a greater understanding of how individual contributions develop and impact subsequent work. This ecosystem is further improved by Lo et al.’s *Semantic Scholar Open Research Corpus* (S2ORC) [10], which offers a large publicly accessible dataset of more than 81 million scholarly articles with full text, metadata and resolved citations that enables a wide range of downstream NLP applications.

Engelmann et al. showed how large language models (LLMs) may be used to make scientific publications easier to understand for readers who aren’t field experts [1]. Their work has shown that LLMs can produce understandable summaries of intricate research, laying the groundwork for science communication outside of academia.

While prior work focuses more on structuring or simplification, our goal is to combine both of these processes to various STEM fields. We propose a model drawing inspiration from Engelmann’s LLM-based simplification approach but targeting only the *novel contributions* of scientific papers. We can isolate the most meaningful innovations of a paper by tagging sentences in a paper’s abstract as a background section or a contribution section, then simplify these contributions into short, plain-language summaries at high-school reading level. In doing so, we hope to connect technical knowledge extraction with inclusive and accessible science communication, helping to make STEM research more available to a wider audience.

3. Approach

3.1. Dataset

In this section, we will provide our systematic process for building a paired corpus of tagged original and high-school-level simplified abstracts to train the simplification module, alongside a comprehensive tagging dataset that pairs a paper’s abstracts with its influential reference abstracts. The first corpus allows for the module to develop the ability of converting intricate multi-sentence inputs into succinct summaries that specialize in highlighting the contributions of a research project. Meanwhile, our tagging dataset enables our tagging module to leverage features such as embeddings and n-gram structures to differentiate between contribution and background sentences. Both resources are drawn from the same source and take advantage of citation contexts to emphasize novel contributions.

3.1.1. Data Source and Citation Retrieval

We begin by processing the Cornell University arXiv dataset from Kaggle, which provides a machine-readable collection of 1.7 million articles, with many relevant features such as titles, abstracts, links to PDFs, and more [11]. Although resourceful, we extract the dataset’s metadata for a list of arXiv IDs and query the Semantic Scholar API to retrieve each paper’s most “influential” references. These citations are selected by a machine-learning model that considers factors such as citation count and the surrounding context (e.g. “Methodology,” “Background” sections) to gauge the impact of each reference on the target paper [12]. Therefore, focusing on these influential predecessors allows us to analyze a given target abstract and outline its contributions given the context of what’s been worked on.

From these sources, we produced a unified data set of 7,000 JSON formatted entries that pairs a target abstract with its influential reference abstract and other key important metadata features as shown in Figure 1.

```

{
  "main": {
    "paperId": "...",
    "title": "...",
    "abstract": "...",
    "...",
  },
  "references": [
    { "paperId": "...", "title": "...", "abstract": "...", "..." }
  ]
}

```

Figure 1: An example of the JSON entry format utilized in our pipeline that shows how each record associates an array of the target article’s most important reference papers and their abstracts with the metadata and abstract of the target paper (under "main").

These entries allow for our tagging module to identify and tag the sentences inside a target abstract. We selected a stratified sample of 100 entries from several fields including biology, physics, and computer science, and carried out a thorough procedure to construct our simplification pairs. For each JSON entry, we read the original abstract, taking note of the general organization in terms of context sentences that explain earlier research or techniques. Afterwards, we identified the distinct sentences that convey the main contributions, which typically explain what was measured, found, or proposed. Subsequently, we then rationalized each possible contribution by looking at the previous abstracts and ensured that it was unique by either introducing a new concept or expanded upon a previous one.

<TASK> Summarize this abstract for high-school audience while highlighting the importance of contributions. </TASK> <ABSTRACT> <BACKGROUND> In multimedia applications, the text and image components in a web document form a pairwise constraint that potentially indicates the same semantic concept. </BACKGROUND> <CONTRIBUTION> We first propose a compound regularization framework to address the pairwise constraint, which can be used as a general platform for developing cross-modal algorithms. </CONTRIBUTION> </ABSTRACT>	<TASK> ... </TASK> <ABSTRACT> ... </ABSTRACT> <SUMMARY> This paper shows how to link text and images by finding shared hidden patterns. The authors build a new mathematical framework and use it to create two tools: one that groups text and images without labels, and another that matches them when labels are present. Their tests prove these tools work better at reducing mistakes and improving accuracy in combining text and image data. </SUMMARY>
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Figure 2: General tagging example illustrating the use of <TASK>, <ABSTRACT>, <BACKGROUND>, <CONTRIBUTION>, and <SUMMARY> tags to highlight semantic roles within an abstract.

For each sampled abstract, we also wrote summaries of 3-5 sentences that specifically highlighted those contributions. We then optimized these summaries to achieve the Flesch Reading Ease target scores of approximately 50-70 for high school students [2]. Next, we drew upon Stanford’s Semantic

Role Labeling framework to organize our tags and improve the interpretability for the model. As shown in Figure 2, the `<TASK>` directly informs the model that it’s a summarization task, prompting it to concentrate on condensing contributions. The precise span of source text that model should be examining is then indicated by the `<ABSTRACT>` tag. While the `<CONTRIBUTION>` tag emphasizes the predicate-argument structures that characterize each work’s novelty, the `<BACKGROUND>` tag is used within the abstract to record contextual setting sentences. This approach adheres to Jurafsky’s “who did what” paradigm, and we can maintain and streamline each paper’s essential contributions by clearly indicating the model with specific semantic instructions [5].

In order to clearly identify the model’s intended output, we lastly surround each human-written summary in a `<SUMMARY>` tag. Through the combination of the JSON metadata, the sentence-level annotations and our curated summary targets, this dataset offers fine tuning for our simplification module and the ability for our tagging module to access influential contexts to distinguish for contributions in a streamlined way.

3.2. Tagging Module

Our tagging module identifies the contribution sentences through a four stage process to automatically filter and label our target abstracts. The first step involves in detecting novelty by looking at n-gram and semantic similarities. Iteratively, we parse each sentence in the target abstract and compare it against the sentences of its most influential predecessors using the JSON entries as aforementioned. We create embeddings with `all-MiniLM-L6-v2`, a sentence embedding model [4]. Subsequently, we compute an n-gram novelty score (fraction of bigrams not seen before) and a semantic similarity score (maximum cosine similarity of all embeddings).

Furthermore, we curated a given threshold for both the novelty and similarity scores as potential contributions. First, we computed the maximum cosine similarity scores for each sentence across our large dataset of abstracts, we plotted the distribution to evaluate where a good cutoff would be. After evaluating, we chose a cutoff at 0.45 because as shown with Figure 3, it sits left to the central mass where most background sentences would be. This allows us to exclude the bulk of highly

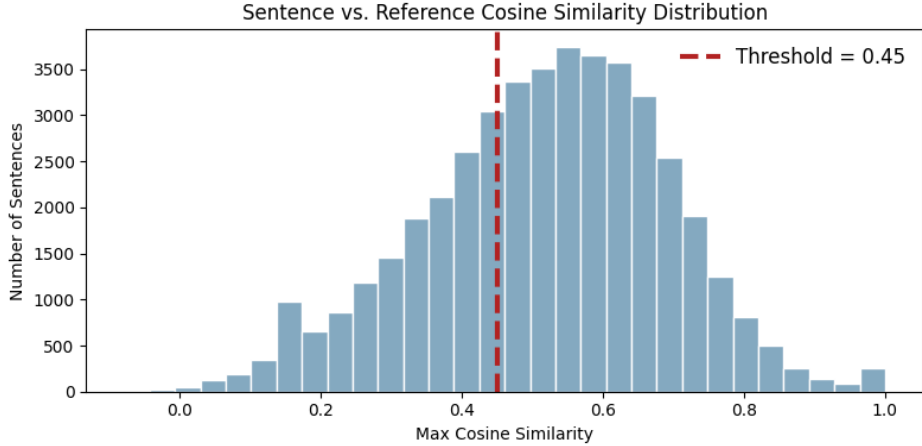


Figure 3: Distribution of maximum cosine-similarity scores between each sentence in the 7,000 arXiv abstracts and its most influential references. The dashed vertical line represents our threshold for potential novel contributions.

redundant text while still allowing a generous amount of the sentences to be considered. All the while, we apply an n-gram novelty filter of 0.3 and above, while testing it with the 100-paper subset, this threshold boosted the accuracy by 5.4 percent compared to lower cutoffs.

Additionally, we apply a positional bias, where we multiple each candidate’s score by $1 + 0.5 \cdot (i/L)$, where i is its zero-based index and L is the total number of sentences. This reflects the tendency for contributions to appear later in the abstracts. Finally, we scan for keyword cues such as “in this paper” and “we propose”, to boost any matching sentence by an additional 30%. We combine these possible contributions and we tag anything that surpasses a threshold of 0.5 as our `<CONTRIBUTION>` tag, with the rest being `<BACKGROUND>` tags.

Ultimately, this tagging module reduces manually tagging and provides accurate input for our downstream simplification model by offering a completely automated technique for identifying and extracting innovative contributions from any scientific abstract.

3.3. Simplification Module

The simplification module leverages a fine-tuned BART-large-cnn model [7] to transform our tagged abstract inputs into concise high-school readable summaries. The transformer-based denoising autoencoder combines bidirectional “understanding” of BERT with auto regressive generation using GPT. The provided checkpoint on huggingface is a publicly released version that’s fine-tuned on the

CNN/DailyMail summarization dataset. To prepare this model, we first pre-process each input with the input tags, cleaning it up for any extra markup. We then tokenize each and we submit them for training. After running finetuning on the dataset with a 70% train, 20% test, and 10% validation, we find parameters of 5 epochs over a small batch size of 4, managing GPU memory limits. We used a learning rate of 2×10^{-5} , and applied a weight decay of 0.05 to help prevent overfitting. At inference time, we generate 30-80 token outputs with beam search (beam size=6, length penalty = 1.0, no-repeat n-gram size =2). This allows for us to have a raw model output that contains a span of text within the <SUMMARY> tags that we can extract as shown in Figure 2. Therefore by combining these contribution labels with a pretrained summarization model, our module consistently produces high-school readable summaries that we can evaluate upon.

4. Evaluation

4.1. Dataset Split

To assess the tagging and summarization modules, we used a stratified sample of 100 JSON entries of different fields as aforementioned. We used 80 entries to fine-tune our BART Seq2Seq model, while the remaining 20 were placed aside as a held-out test set. This train-test split ensures that our assessment takes into account of generalization to unseen abstracts and reference contexts.

4.2. Metrics

For this section, we'd like to not only capture the semantic fidelity and structural accuracy of both the tagging and summarization models, but also the overall readability of our outputs. As a result, we adopt four complementary metrics as shown in Table 1.

4.2.1. BERTScore

BERTScore is an evaluation metric that measures similarity between two pieces of text by leveraging contextual embeddings from pre-trained transformers like BERT. BERTScore computes token-level similarity using cosine similarity between embeddings, allowing it to capture deep semantic

Evaluation Metric	Formula
BERTScore	$\text{BERTScore}(C, S) = \frac{1}{ C } \sum_{i=1}^{ C } \max_j \cos(f(C_i), f(S_j))$
ROUGE	$\text{ROUGE-n}_{\text{recall}} = \frac{ \text{n-grams}_{\text{cand}} \cap \text{n-grams}_{\text{ref}} }{ \text{n-grams}_{\text{ref}} }, \text{ROUGE-n}_{\text{prec}} = \frac{ \text{n-grams}_{\text{cand}} \cap \text{n-grams}_{\text{ref}} }{ \text{n-grams}_{\text{cand}} },$ $\text{ROUGE-L} = \frac{\text{LCS}(C, S)}{ S }, \text{ROUGE-L}_{\text{Sum}} = \frac{1}{N} \sum_{k=1}^N \text{ROUGE-L}(C_k, S_k)$
Tagging Accuracy	$\text{Acc}_{\text{tag}} = \frac{\#\{\text{correctly tagged sentences}\}}{\#\{\text{total sentences}\}}$
Flesch Reading Ease	$\text{FRE} = 206.835 - 1.015 \frac{\text{words}}{\text{sentences}} - 84.6 \frac{\text{syllables}}{\text{words}}$

Table 1: Formulas for our evaluation metrics: BERTScore, ROUGE (1, 2, L, L Sum), Tagging Accuracy, and Flesch Reading Ease.

alignments even when surface forms differ [13]. According to Zhang et al., these scores correlate better with human judgement across various tasks, including machine translation.

4.2.2. ROUGE Score

ROUGE-1 & ROUGE-2. These metrics measure overlap of unigrams (ROUGE-1) and bigrams (ROUGE-2) between the candidate and reference summaries. ROUGE-1 captures word-level coverage, while ROUGE-2 evaluates the local phrase fluency and coherence [8].

ROUGE-L & ROUGE-Lsum. ROUGE-L computes the longest common subsequence (LCS) between candidate and references to assess the sentence-level structure. ROUGE-Lsum averages these LCS scores across sentences for multi-sentence coherence. ROUGE-L should reflect fluency and grammar-proficiency of summaries, while Lsum evaluates coherence at nuanced level [3].

4.2.3. Tagging Accuracy

This score measures the proportion of sentences for which the tagging module correctly predicts the <BACKGROUND> or <CONTRIBUTION> tags. High tagging accuracy would indicate that we have a reliable identification of contribution vs. background sentences, which is crucial for guiding our simplification model [5].

4.2.4. Flesch Reading Ease

The Flesch Reading Ease (FRE) score evaluates the readability of a text based on the average sentence length and syllable density. Score ranges from 0 (very difficult) to 100 (very easy), where a score of 60-70 corresponds to a standard high-school reading level, and 50-60 is a more advanced

high-school reading level (seniors).

4.3. Experimental Setup

To quantify the contributions of both tagging and summarization stages, we have developed four systems on our held-out evaluation set. In the first model, we run the raw abstracts through our tagging module alone and measure the tagging accuracy against the ideal tagged abstracts. The subsequent models are set up to comparatively assess the performance of the fine-tuned Seq2Seq model.

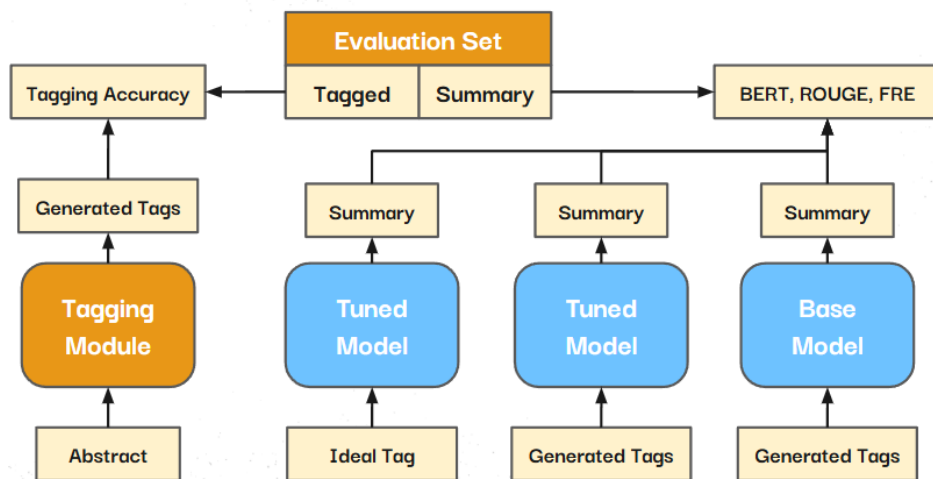


Figure 4: Overview of our evaluation pipeline: abstracts are tagged, and both ideal and generated tags feed into summarization models (fine-tuned and baseline). We then measure Tagging Accuracy and evaluate summaries with BERTScore, ROUGE, and Flesch Reading Ease.

As shown in Figure 4, we use the ideal tags into our fine-tuned model, to establish an upper bound on the summary quality when we have precise tagging. In the third system, we replace the ideal tags with the ones generated from the tagging module, allowing us to see how potential tagging errors would propagate to the performance of generating summaries. Following this, we use the regular BART model with the set of generated tags as a baseline output to show the improvements from the fine-tuned model. For these systems indicated in blue, we generate summaries for all 20 held-out abstracts and evaluate them using BERTScore for semantic fidelity, and then ROUGE-1/2/L/Lsum for lexical and structural overlap. In the final step, we look at the Flesch Reading Ease to assess the readability and ensure that the model is simplifying the abstracts.

5. Results

5.1. Tagging Module

To evaluate the performance of our contribution tagging system, we applied this module to the held-out subset of 20 abstracts. Due to a minor preprocessing bug that corrupted 4 of the entries, we will report the results on the remaining 16 valid abstracts.

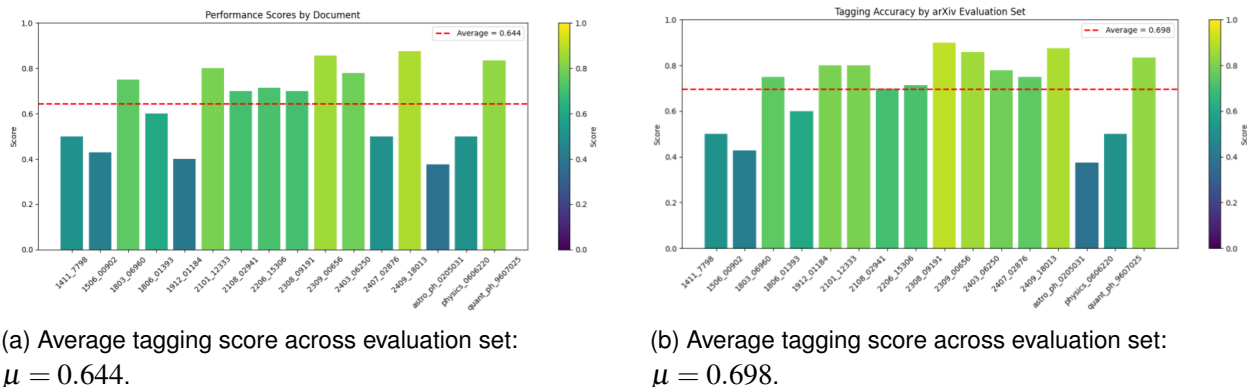


Figure 5: Comparison of model behavior before and after raising the tagging threshold from 0.30 to 0.45. (a) shows raw performance scores by document at the original threshold; (b) shows tagging accuracy once the threshold has been fine-tuned.

As shown in Figure 5b, our tagging module achieved an average tagging accuracy of 0.698 across the 16 samples. Individual documents scores ranged from as low as 0.37 to as high as 0.91, with a higher concentration around 0.7-0.85. Abstracts 2309.09191, 2309.00656, and 2409.18013 had the highest accuracy, indicating that their contributions were semantically and structurally unique. Abstracts such as `physics.0606220`, and `astro.ph.0205031`, on the other hand, demonstrated lower accuracy, most likely due to their less conventional abstract structures or a possible reliance on background context.

The tagging module’s efficacy is derived from a combination of n-gram novelty filtering, semantic similarity, positional bias, and cue phrase identification. Our empirical analysis aforementioned, led to us choosing a more acceptable threshold to improve the performance of our tagging module. As depicted from Figure 5a, where we improve the overall performance by 5.4% on our development

subset. Furthermore, our positional weighting and keyword boosting algorithms were important in filtering out a number of contributions, but we may explore robust classification models to tailor it to given STEM fields (i.e. astrophysics and physics) and provide more keywords to improve the accuracy.

5.2. Simplification Module

We evaluate three systems on our held-out test set of 20 abstracts: 1) Ideal Tagging: fine-tuned BART with manual contribution tags, 2) Automatic Tagging: fine-tuned BART with automatically generated tags from our tagging module, and 3) Default Output: off-the-shelf BART without any finetuning or tagging.

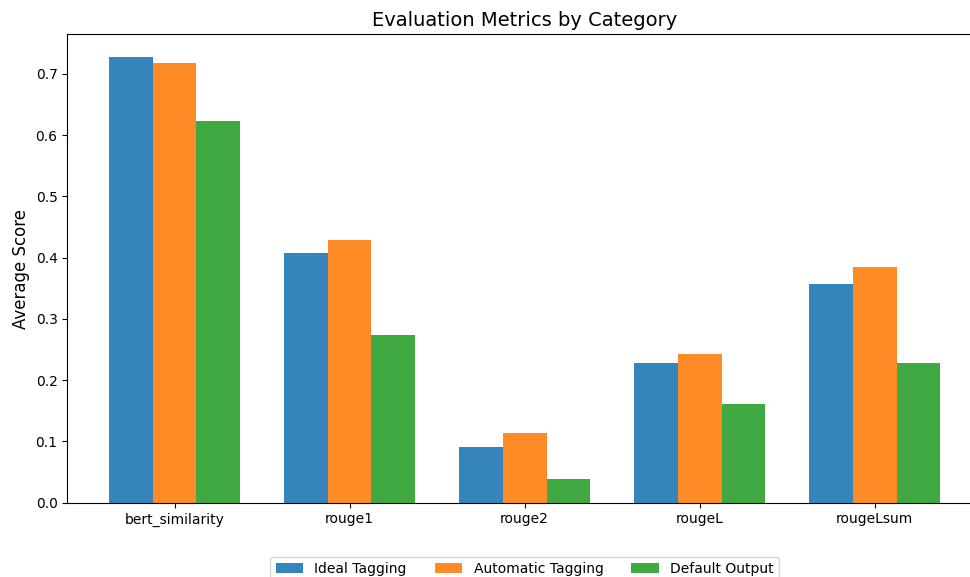


Figure 6: Average evaluation metrics for the three systems: Ideal Tagging (manual tags), Automatic Tagging (our tagging module), and Default Output (base BART). The results show strong gains in BERTScore and ROUGE when using contribution tagging, with automatic tagging performing nearly as well as the ideal upper bound.

As seen in Figure 6, we see that tagging generally improves summary quality. Both Ideal Tagging and Automatic Tagging outperform the Default Output across all metrics. For example, BERTScore improves from 0.62 (Default) to 0.73 (Ideal) and 0.72 (Automatic), which is about a 17% relative gain. ROUGE-1 improves from 0.28 to over 0.4, and ROUGE-2 nearly triples. The Automatic Tagging system performs almost as well as the Ideal Tagging system, and even slightly better on

some ROUGE metrics. For example, ROUGE-1 improves from 0.41 to 0.43 on Ideal to Automatic, and ROUGE-2 improves from 0.09 to 0.11. We see similar improvements in ROUGE-L and ROUGE-Lsum, improving from 0.23 to 0.24, and improving from 0.36 to 0.38 respectively.

5.3. Flesch Results

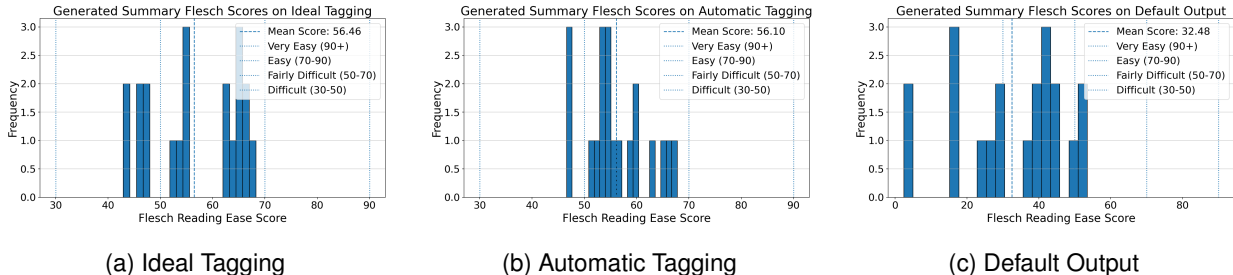


Figure 7: Flesch Reading Ease score distributions for summaries generated under Ideal Tagging, Automatic Tagging, and Default Output. Both tagging systems produce summaries within the high-school readability range (50–70), while the default system remains too difficult for non-expert readers.

As seen in Figure 7, the baseline BART model without any tagging generated summaries with an average Flesch Reading Ease score of 32.48, which falls into the “difficult” category, which is too complex for high school readers as established earlier. In contrast, both the Ideal Tagging system (with manual tags) and the Automatic Tagging system (which uses our tagging module) achieved mean Flesch Reading Ease scores of 56.46 and 56.10, respectively. This places those into the correct high-school level reading bin. The histograms in Figure 7 show most summaries clustered between 50 and 60, where the default output was mostly scattered, and many falling below 30.

6. Conclusion

We have introduced a two-stage approach for making scientific abstracts more accessible: first, we automatically tag and identify the key “contribution” sentences in an abstract, and second, by transforming these sentences into high-school level summaries. Our tagging modules combines novelty detection, positional bias, and cue-phrase signals, which allows us to distinguish between background and contribution statements while our fine-tuned BART-based simplification module produces concise paraphrases that preserve the paper’s primary contributions. Our evaluation on

a held-out test set shows that our tagging module at least matches and improves upon manually tagged inputs, with substantial gains in both semantic fidelity like BERTScore and lexical overlap with ROUGE compared to an untargeted baseline model. Additionally, we improve from the baseline and achieve similar results to the Ideal tagging in Flesch Reading Ease score, showing a near 72% improvement. Together, the results show that targeted contribution tagging is not only faster than manual tagging, but of equal or better quality. Additionally, our methodology of targeted contribution tagging can significantly improve focus and readability of generated scientific summaries, allowing for a more inclusive science communication and easier access to technical fields.

7. Future Work

While our work focuses primarily on STEM papers, we hope this work can be expanded domain-wise to include coverage beyond the sciences, including social sciences, humanities, and interdisciplinary research, as the summaries may be more interpretable. Recruiting undergraduate students or human experts in the loop to manually tag the abstracts and write high-quality summaries could improve the quality of the our dataset, especially if there were domain experts.

Furthermore, integrating a “layman dictionary” or a controlled vocabulary could automate the replacement of technical words with a simplified version of these words, making the texts easier to read as well. On the modeling side, other architectures for Seq2Seq like T5 or Pegasus should be evaluated to identify the most effective model. Beyond model improvements, future works include adapting the system to multilingual inputs and outputs to increase the accessibility to those who do not speak primarily English, and also enhancing novelty detection through advance language models or graph-based techniques.

References

- [1] B. Engelmann, C. K. Kreutz, F. Haak, and P. Schaer, “ARTS: Assessing readability & text simplicity,” in *Findings of the Association for Computational Linguistics: EMNLP 2024*, Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 14 925–14 942. [Online]. Available: <https://aclanthology.org/2024.findings-emnlp.877/>
- [2] R. Flesch, “A new readability yardstick,” *Journal of Applied Psychology*, vol. 32, no. 3, pp. 221–233, 1948.
- [3] N. Glazkov and I. Makarov, “Utterance-Aware Adaptive Data Labeling and Summarization: Exploiting Large Language Models for Unbiased Dialog Annotation,” *IEEE Access*, vol. 12, pp. 150 793–150 806, 2024, publisher: Institute of Electrical and Electronics Engineers (IEEE). [Online]. Available: <https://ieeexplore.ieee.org/document/10711180/>
- [4] Hugging Face, “sentence-transformers/all-minilm-l6-v2,” <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>, 2021, accessed: 2025-05-01.
- [5] D. Jurafsky and J. H. Martin, “Semantic role labeling,” in *Speech and Language Processing (3rd ed.)*. Draft Chapter, Stanford University, 2021. [Online]. Available: https://web.stanford.edu/~jurafsky/slp3/old_sep21/19.pdf
- [6] D. Jurgens, S. Kumar, R. Hoover, D. McFarland, and D. Jurafsky, “Measuring the evolution of a scientific field through citation frames,” in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2018, pp. 148–157. [Online]. Available: <https://aclanthology.org/N18-2014>
- [7] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, O. Levy, V. Stoyanov, and L. Zettlemoyer, “Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 7871–7880.
- [8] C.-Y. Lin, “ROUGE: A package for automatic evaluation of summaries,” in *Text Summarization Branches Out*. Association for Computational Linguistics, 2004.
- [9] M.-H. Liu, A.-Z. Yen, H.-H. Huang, and H.-H. Chen, “ContributionSum: Generating Disentangled Contributions for Scientific Papers,” in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*. Birmingham United Kingdom: ACM, Oct. 2023, pp. 5351–5355. [Online]. Available: <https://dl.acm.org/doi/10.1145/3583780.3615115>
- [10] K. Lo, L. L. Wang, M. Neumann, R. Kinney, and D. Weld, “S2ORC: The semantic scholar open research corpus,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds. Online: Association for Computational Linguistics, Jul. 2020, pp. 4969–4983. [Online]. Available: <https://aclanthology.org/2020.acl-main.447/>
- [11] C. University, “Kaggle: arXiv Dataset,” <https://www.kaggle.com/datasets/Cornell-University/arxiv>, 2025.
- [12] M. Valenzuela, V. Ha, and O. Etzioni, “Identifying meaningful citations,” in *Proceedings of the “Scholarly Big Data: AI Perspectives, Challenges, and Ideas” Workshop at the Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015. [Online]. Available: <http://ai2-website.s3.amazonaws.com/publications/ValenzuelaHaMeaningfulCitations.pdf>
- [13] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019.

8. Appendix

Complete code for our project is detailed in the GitHub repository, including execution instructions.

The script for creating the JSON dataset is also provided within the repository and access to the reference dataset is located in the repository as well.

https://github.com/javierlinero/tra301_final